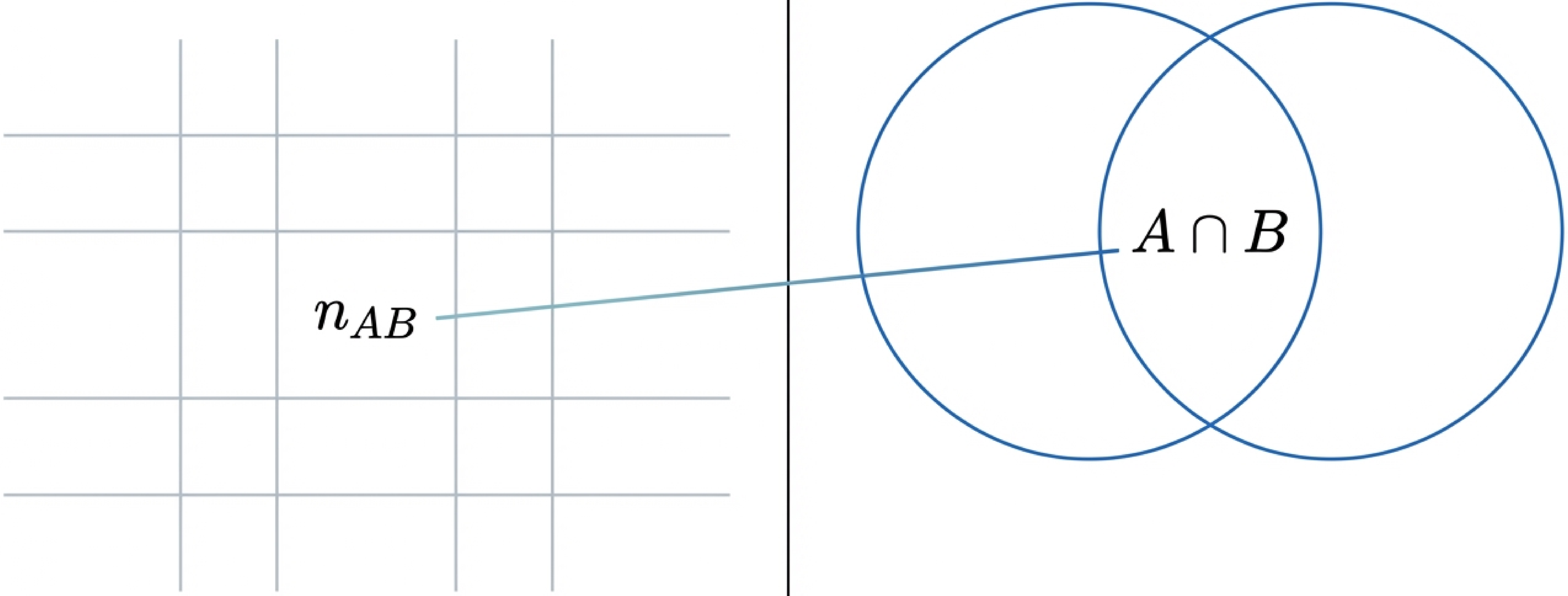
Probability & Data Structure

From Contingency Tables to Venn Diagrams

A structural toolkit for organising, measuring, and visualising categorical data relationships.



Data Science Methodologies | Exploratory Data Analysis

The Analyst’s Toolkit

To unlock insights in categorical data, we follow a progressive workflow. This deck connects the mathematical structure of tables to the intuitive nature of visualisations.

Organise
Contingency Tables

Measure
Marginal & Joint Probability

Visualise
Venn Diagrams

Compute
Python Implementation

Topic 1: The Contingency Table

A contingency table (or cross-tabulation) displays the frequency distribution of variables. It is the foundational structure for calculating joint, marginal, and conditional probabilities.

	Event B	Event B’
Event A		
Event A’		

The 2x2 Matrix:

Rows represent outcomes of Variable 1, Columns represent outcomes of Variable 2. The intersection cells hold the frequency of simultaneous occurrences.

Deconstructing the Anatomy

The Intersection: Number of outcomes where A and B occur simultaneously.

	Event B	Event B'	Total
Event A	n_{AB}	$n_{AB'}$	n_A
Event A'	$n_{A'B}$	$n_{A'B'}$	$n_{A'}$
Total	n_B	$n_{B'}$	N

The Margins: Marginal totals for individual events.

The Margins: Marginal totals for individual events.

The Grand Total: Sum of all observations.

Topic 2: Marginal Probability

The probability of a single event occurring, regardless of other events. It is calculated by summing along the rows or columns—values **literally found in the margins** of the table.

$$P(A) = \frac{n_A}{N}$$

$$P(\text{USA}) = \frac{300}{1000} = 0.3$$

User Distribution (USA vs UK)

	Premium	Basic	Total
USA	80	220	**300**
UK	120	580	**700**
Total	**200**	**800**	**1000**

$$P(\text{Premium}) = \frac{200}{1000} = 0.2$$

Topic 3: Joint Probability

Definition: The probability that two events happen at the same time (the intersection).

$$P(A \cap B) = \frac{n_{AB}}{N}$$

User Distribution (USA vs UK)

	Premium	Basic	Total
USA	**80**	220	**300**
UK	120	580	**700**
Total	**200**	**800**	**1000**

$$P(\text{USA} \cap \text{Premium}) = \frac{80}{1000} = 0.08$$

Note: If events A and B are independent, $P(A \cap B) = P(A) \times P(B)$.

Deep Dive: Conditional Probability

Shrinking the Sample Space

Probability that A occurs given B occurred. This effectively shrinks the sample space from N (Grand Total) to n_B (the Marginal Total of B).

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{n_{AB}}{n_B}$$

$$P(B|A) = \frac{n_{AB}}{n_A}$$

User Distribution (USA vs UK)

	Premium	Basic	Total
USA	80	220	300
UK	120	580	700
Total	200	800	1000

When calculating P(USA | Premium), we ignore the Basic users. The denominator becomes the Premium Total (200), not the Grand Total (1000).

Case Study: Website Analytics

Scenario: Analysing the relationship between clicking an ad and making a purchase.

User Behavior (Ad Click vs Purchase)

	Purchased	Did Not Purchase	Total
Clicked Ad	200	300	**500**
Didn't Click	100	400	**500**
Total	**300**	**700**	**1000**

Joint Probability (Likelihood of doing both):

$$P(\text{Clicked} \cap \text{Purchased}) = \frac{200}{1000} = 0.2$$

Marginal Probability (Likelihood of clicking):

$$P(\text{Clicked}) = \frac{500}{1000} = 0.5$$

Conditional Probability (Effectiveness of the ad):

$$P(\text{Clicked} \mid \text{Purchased}) = \frac{200}{300} \approx 0.67$$

Interpretation: While only 20% of all visitors did both, 67% of purchasers had clicked the ad.

Implementation: Python for 2x2 Tables

Using `pandas` to create the dataframe and derive probabilities.

```
import pandas as pd

# Create contingency table
data = { "Purchased": [200, 100], "Did_Not_Purchase": [300, 400] }
df = pd.DataFrame(data, index=["Clicked", "Did_Not_Click"])

# Calculate Totals
df["Total"] = df.sum(axis=1)
df.loc["Total"] = df.sum()
N = df.loc["Total", "Total"]
```

Retrieves the denominator N (1000)

```
# Calculate Probabilities
p_clicked_purchased = df.loc["Clicked", "Purchased"] / N
p_clicked = df.loc["Clicked", "Total"] / N

print(f"P(Clicked ∩ Purchased): {p_clicked_purchased:.2f}")
```

Expansion: The 3x3 Table

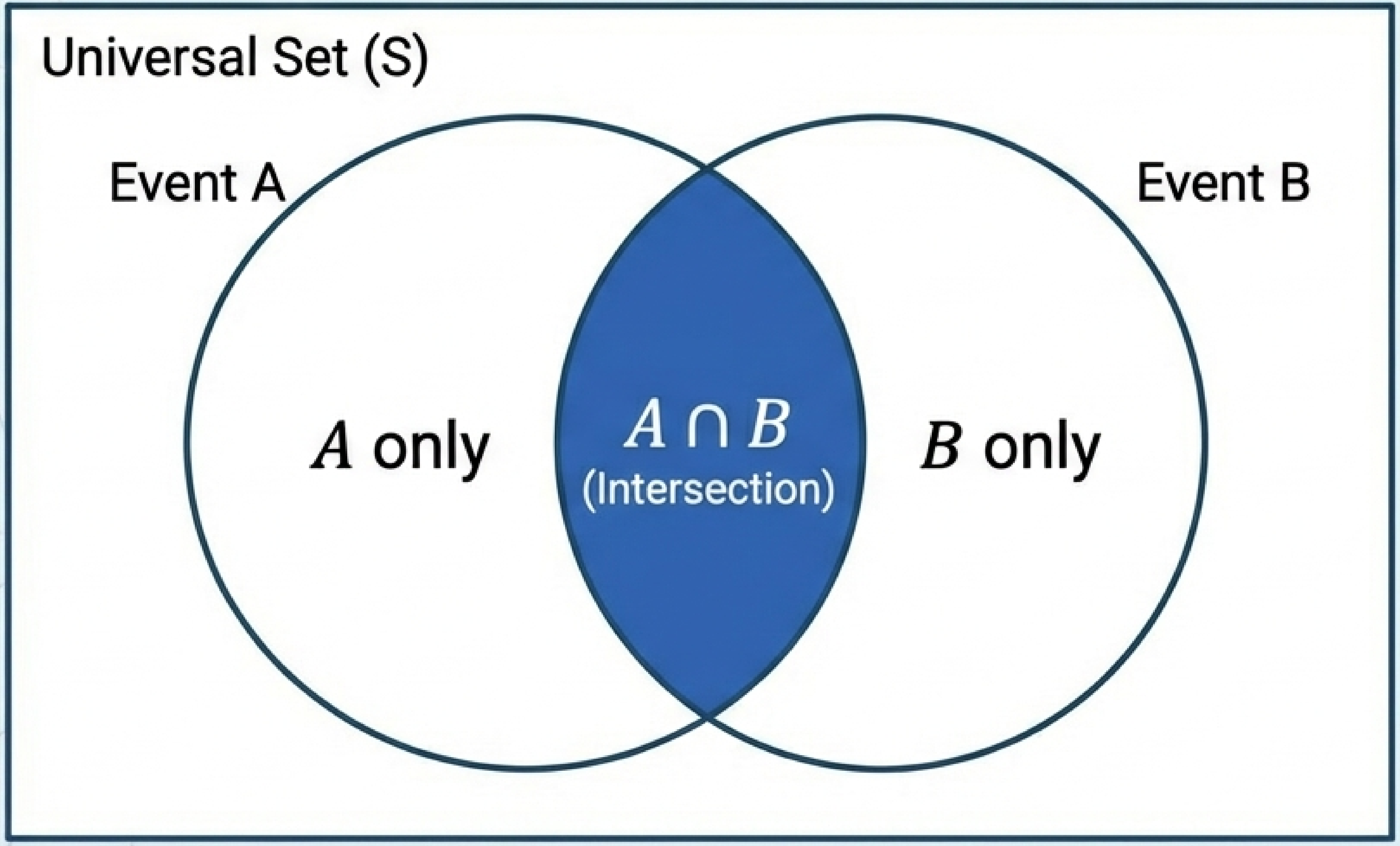
Real-world data is rarely just binary. Here we analyse 'Programming Language' vs. 'Experience' to see how the logic scales.

	Python	R	Java	Total
Beginner	40	10	20	**70**
Intermediate	35	25	15	**75**
Expert	25	15	15	**55**
Total	**100**	**50**	**50**	**200**

- **Marginal:** $P(\text{Python}) = \frac{100}{200} = 0.5$ (Sum of Column)
- **Joint:** $P(\text{Beginner} \cap \text{Python}) = \frac{40}{200} = 0.2$ (Intersection Cell)
- **Conditional:** $P(\text{Python} \mid \text{Beginner}) = \frac{40}{70} \approx 0.57$ (Row Total as denominator)

Topic 4: Venn Diagrams

A graphical representation of sets and their relationships. In probability, it visualises events as sets and shaded regions as probabilities.

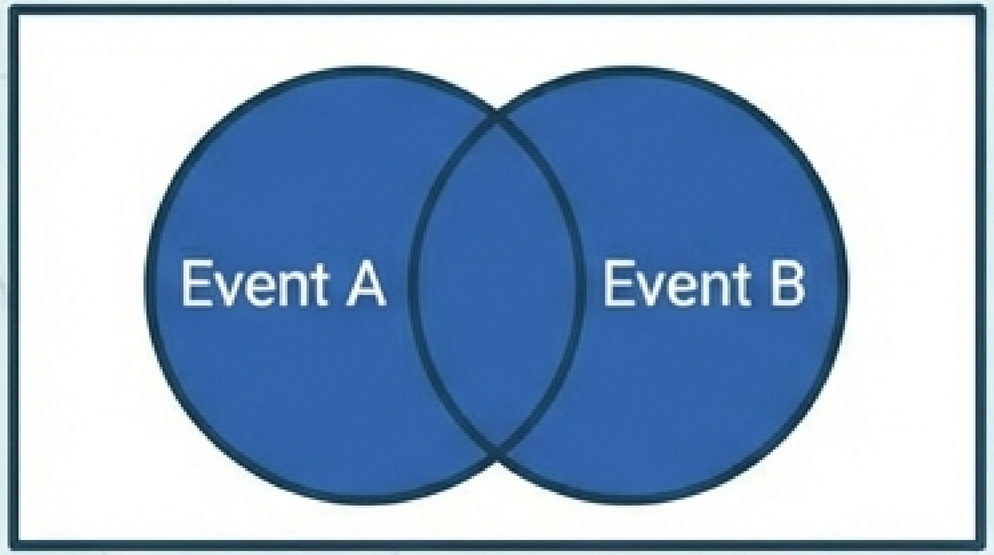


Essential for **Exploratory Data Analysis (EDA)** to visualise categorical overlaps, such as customers using Mobile vs. Desktop.

Venn Logic & Regions

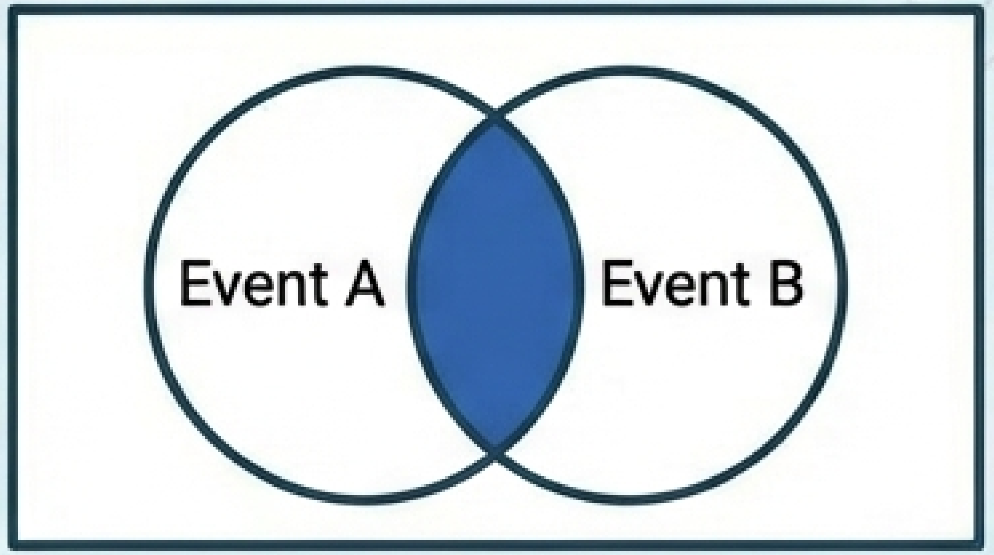
Union ($A \cup B$)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



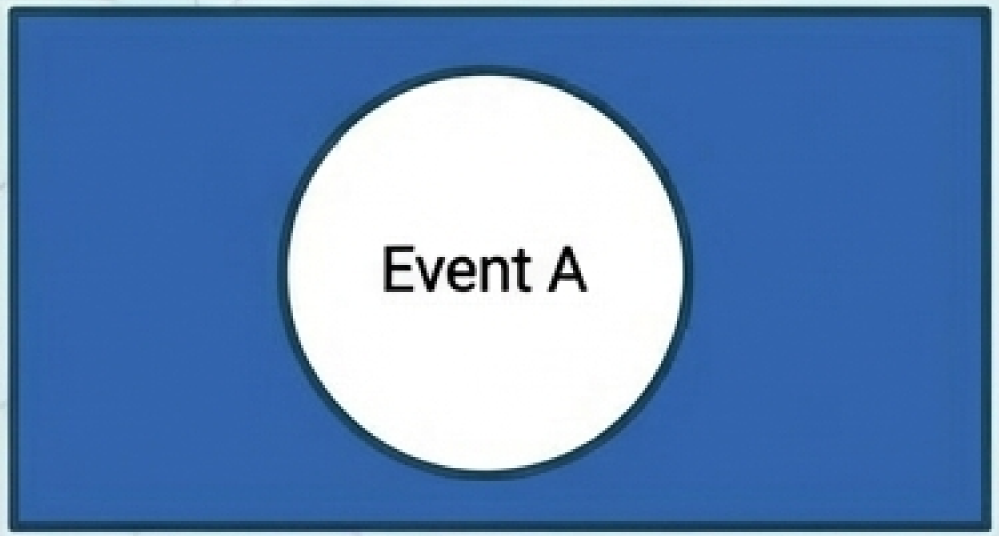
Intersection ($A \cap B$)

$$P(A \cap B) = P(A) \times P(B) \text{ (if independent)}$$



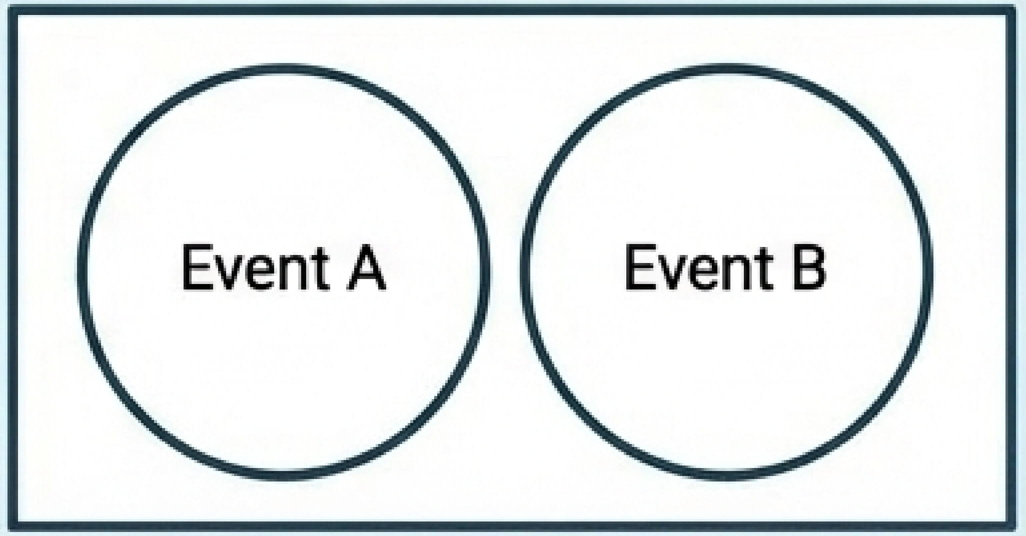
Complement (A')

$$P(A') = 1 - P(A)$$



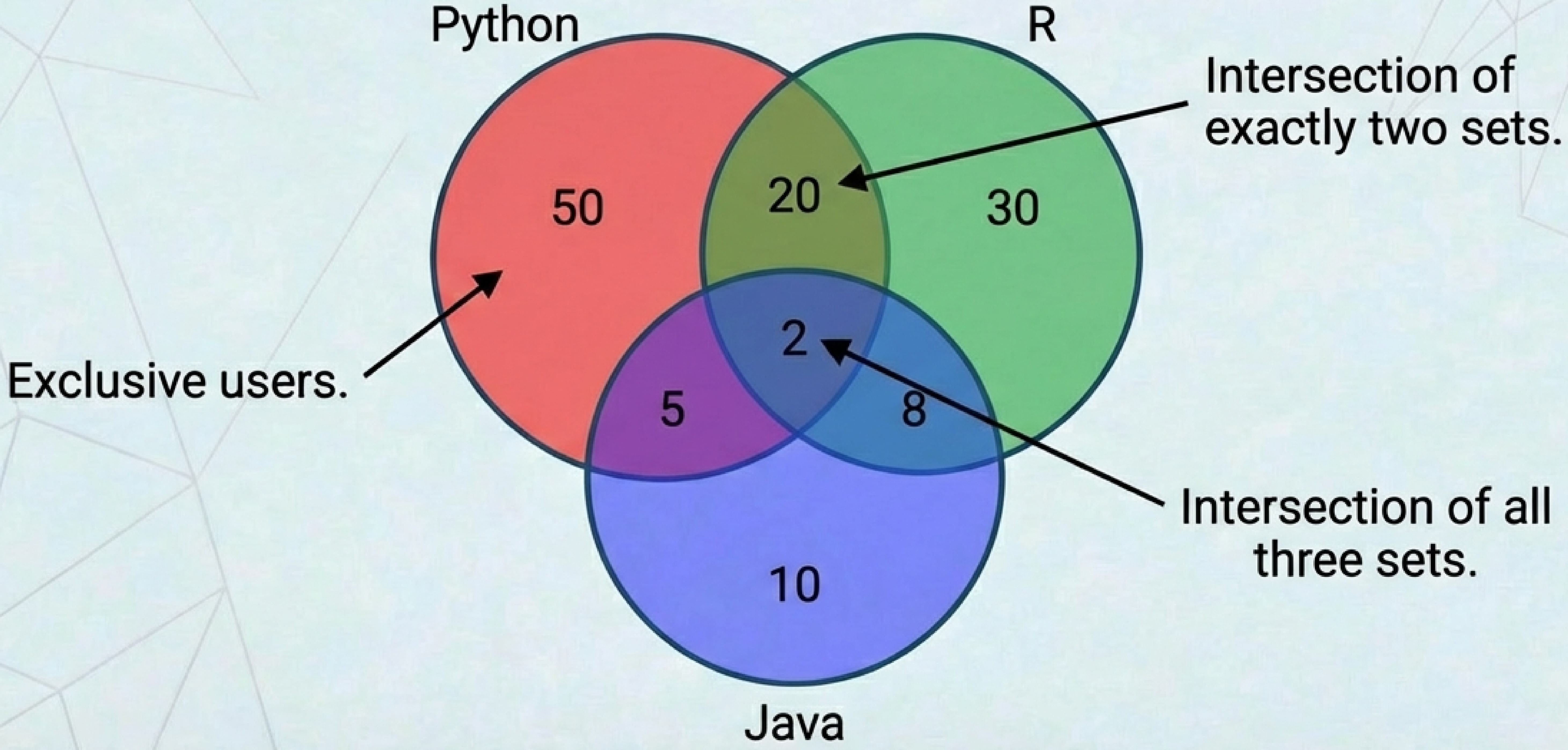
Mutually Exclusive

$$P(A \cap B) = 0$$



Visualising 3-Variable Complexity

Mapping the Programming Language Data



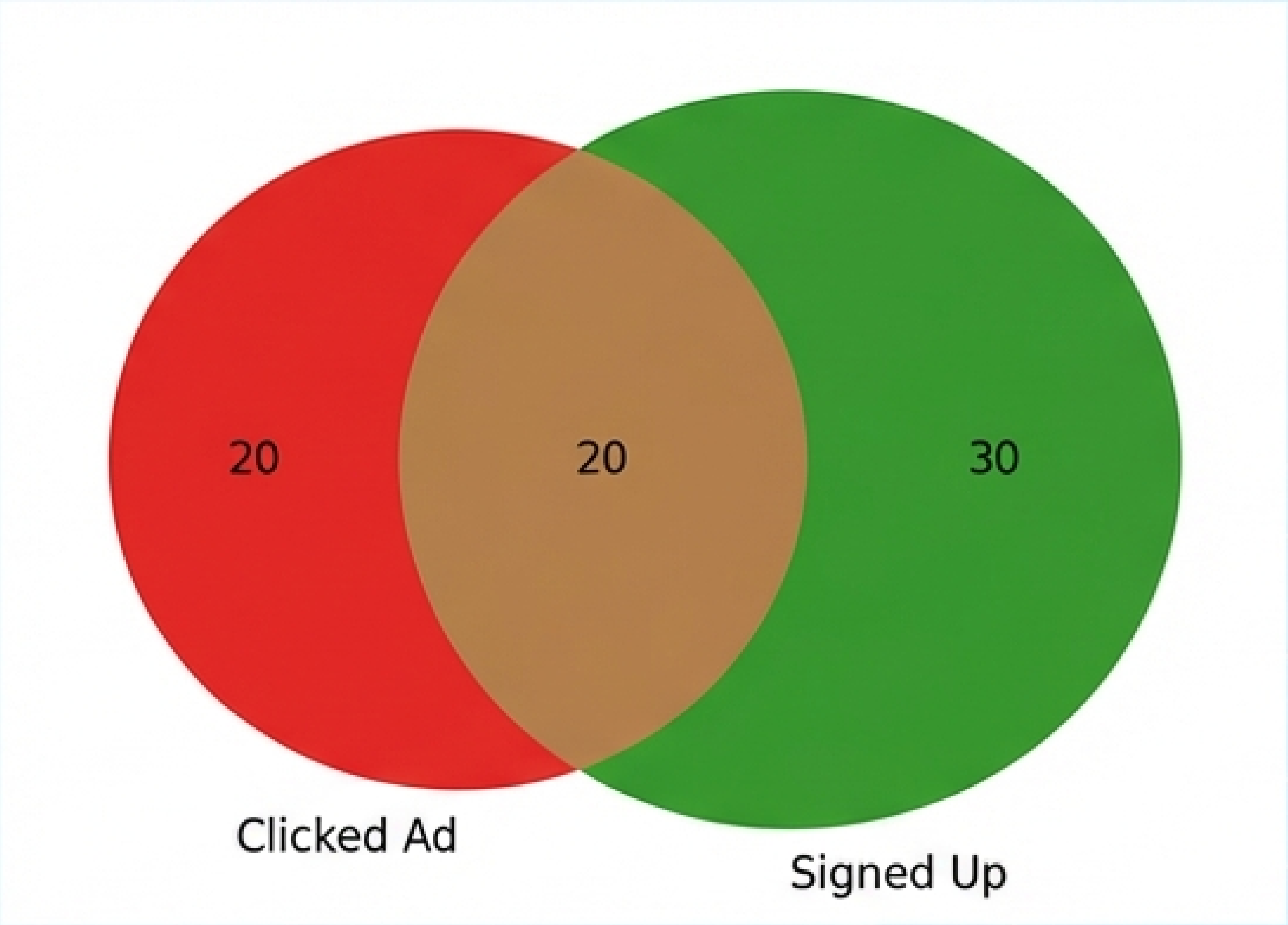
Implementation: Python for Venn Diagrams

Using **`matplotlib_venn`**. Note: The function requires defining exclusive regions by subtracting the intersection.

```python
from matplotlib_venn import venn2
import matplotlib.pyplot as plt
import matplotlib.pyplot as plt

P_A = 40   # Clicked Ad
P_B = 50   # Signed Up
P_AB = 20  # Did Both

# Note: Subtract intersection to get 'exclusive' regions
venn2(subsets=(P_A - P_AB, P_B - P_AB, P_AB),
      set_labels=('Clicked Ad', 'Signed Up'))
plt.show()
```



Summary & Application

Key Takeaways

Structure

Contingency tables organise raw frequency distributions (n_{AB}).

Measurement

Marginal and Joint probabilities quantify the relationships within that structure.

Visualisation

Venn diagrams provide an intuitive geometric view of unions and intersections.

Why this matters for Data Science

- **Feature Analysis:** Determining if variables are correlated (e.g., Chi-square tests).
- **Bayesian Statistics:** $P(A)$ (Marginal) is often the denominator (Evidence) in Bayes' theorem.
- **Classification:** Essential for Naïve Bayes algorithms and association rules.